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Dissemination of antimicrobial resistance in *Klebsiella* spp. from urban aquatic environments: a multi-country genomic perspective

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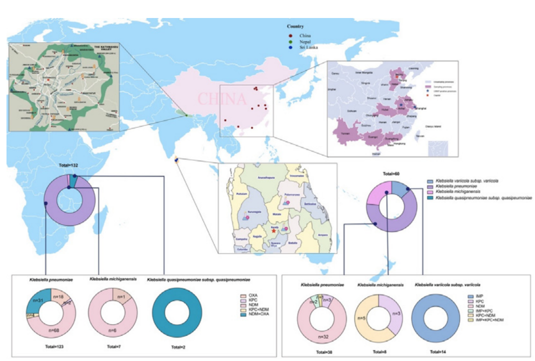
HIGHLIGHTS

- Significant disparities in CRKP prevalence found between China and Nepal's urban waters.
- CRKP isolates exhibit high resistance, especially *qnr* and *tmexCD-toprJ* genes in Nepal.
- High-virulence CRKP strains in Nepal confirmed via *Galleria mellonella* model.
- Geographic patterns show distinct CRKP resistance profiles in China and Nepal.
- Urban waters act as reservoirs for resistance genes, urging integrated management.

GRAPHICAL ABSTRACT

Enhanced representation of sampling locations and prevalence of carbapenem-resistant Klebsiella spp. in urban aquatic environments of China, Nepal, and Sri Lanka. Regions are color-coded by sampling prevalence, with circular charts detailing Klebsiella species and carbapenemase types. Wastewater management practices are visualized for hospitals to correlate with ARG dissemination patterns.



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ABSTRACT

Introduction: Antibiotic resistance, particularly carbapenem-resistant *Klebsiella pneumoniae* (CRKP), poses significant clinical and environmental threats, especially in urban aquatic ecosystems and hospital wastewaters.
Objectives: This study aims to analyze the epidemiological and genomic features of CRKP isolates in urban aquatic environments and evaluate their public health and environmental impacts.
Methods and Results: Water samples were collected from 113 rivers and 3 hospitals in China, Sri Lanka, and Nepal to isolate carbapenem-resistant *Klebsiella* spp. isolates. Antimicrobial susceptibility testing, whole-genome sequencing, and bioinformatics analyses were performed to characterize resistance phenotypes, antibiotic resistance genes (ARGs), and evolutionary trends. Big data analysis further elucidated

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Urban aquatic environments
Wastewater treatment

the genomic characteristics of CRKP in global water sources, and *Galleria mellonella* larvae were used to assess virulence. Statistical analysis validated the findings. A total of 192 carbapenem-resistant *Klebsiella* spp. isolates were identified from urban aquatic ecosystems in China (n = 60) and Nepal (n = 132), with CRKP (n = 161) being the predominant species. All CRKP isolates exhibited a multidrug-resistant phenotype, yet significant differences in resistance profiles and associated ARGs were observed between isolates from the two countries. Nine carbapenem resistance genes (CRGs) were detected, with bla_{NDM-1} being the most prevalent (57.8 %). Correlation analysis revealed a strong association between these CRGs and multiple Inc-type plasmids. Global genomic analysis of CRKP from water sources across eight countries identified ten distinct CRGs across 45 serotypes, with KL64 being the most predominant. Notably, carbapenem-resistant hypervirulent *Klebsiella pneumoniae* was detected in water samples from Nepal.

Conclusion: Our findings highlight significant regional disparities in CRKP prevalence and ARG dissemination across urban aquatic environments, with Nepal showing the highest prevalence, particularly in untreated rivers. China exhibited lower prevalence but distinct resistance gene profiles, while no CRKP was detected in Sri Lanka, underscoring the impact of environmental management and healthcare infrastructure on ARG spread.

Introduction

Carbapenem-resistant *Klebsiella pneumoniae* (CRKP), a critical priority pathogen on the World Health Organisation List, has evolved from a nosocomial threat to a cause of severe community-acquired infections [1–3]. *K. pneumoniae* resistance to extended-spectrum β-lactamases (ESBLs) and carbapenemases, along with its role in multidrug resistance (MDR), makes it a key member of the ESKAPE pathogens [4]. Its global impact of CRKP is evident, with over 90,000 infections and 7,000 deaths annually in Europe alone, and high mortality rates in Saudi Arabia (32.4 %) and China (15.8 %) [5–7]. A further study has indicated that the occurrence of CRKP-BSI increased from 20.69 to 37.40 % from 2012 to 2019, and the overall 28-day mortality rates of CRKP-BSI patients were significantly higher than that of CSKP-BSI ($P < 0.001$) [8]. Globally, the trend of increasing MDR rates is alarming [9], with some regions reporting over 75 % of *K. pneumoniae* bloodstream infections as MDR [10].

Consequently, the spread of CRKP represents a global concern that extends beyond healthcare settings, posing a significant global concern for community environments, particularly within aquatic ecosystems. Rivers, as integral components of the environment, not only serve as crucial resources for human consumption and agriculture but also act as potential reservoirs and conduits for the dissemination of antibiotic-resistant bacteria. Apart from antimicrobial resistance, another worrisome development is related to the evolution of carbapenem-resistant, hypervirulent *K. pneumoniae* (CR-hvKp). The first CR-hvKp was reported from clinical settings with fatal infections in 2015 [11]. Carrying various virulence genes including the aerobactin synthesis operon (*iuc*ABCD), the outer membrane ferric aerobactin receptor (*iut*A), the salmochelin siderophore related genes (*iroBCDN*), and the regulator of mucoid phenotype A (*rmpA* and/or *rmpA2*) have been recognized as molecular indicators of hvKp. The detection of CR-hvKp in hospital and urban aquatic environments, especially rivers, raises significant concerns as it can contaminate water sources and spread ARGs via clonal and horizontal gene transfer, worsening the antimicrobial resistance crisis [12].

The Himalayas, known as the "Asian Water Towers," and Sri Lanka's river systems are essential water sources for millions [13,14]. The interconnected river networks of China, Nepal, and the distinct yet crucial water systems of Sri Lanka offer a novel perspective for comparative studies on the prevalence, resistance mechanisms, and virulence traits of aquatic CRKP. At present, several studies around the world has focused the emergence of CRKP and CR-hvKp in different environments including hospital wastewater and urban wastewater [15,16]. However, research on their comparative occurrence and transmission dynamics in interconnected aquatic systems, particularly between neighboring countries sharing the same watershed, remains scarce.

In this study, a large-scale CRKP screening of urban aquatic environment samples was carried out in several provinces in the three countries China, Sri Lanka and Nepal. We revealed the epidemiological distribution and genomic characteristics of CRKPs in urban water environments across different countries. Our findings provide a comprehensive analysis of the antimicrobial resistance, virulence, and evolutionary traits of CRKP isolates from urban water sources in China and Nepal. Furthermore, through extensive big data analysis, we elucidated the genomic epidemiology and evolutionary dynamics of CRKP in global water systems. This study offers valuable insights into the molecular characteristics of CRKP in aquatic environments across diverse geographical regions, highlighting the need for continuous surveillance to mitigate its potential public health and environmental risks.

Methods

Sample collection and bacterial isolation

In September and October 2023, sampling was conducted from 10 cities in China (Hefei, Anhui; Beijing; Nanning, Guangxi; Baodin, Hebei; Wuhan, Hubei; Suzhou, Jiangsu; Shanghai; Shenzhen, Guangdong; Tianjin; Kunming, Yunnan), and three cities of Polonnaruwa, Kurunegala, and Kandy in Sri Lanka, together with Kathmandu City in Nepal. Three distinct sources of samples were collected, including urban aquatic, hospital wastewater, and drinking water source lake samples. A total of 113 river and 3 hospitals were sampled. Two samples were collected from each river and each hospital wastewater, with one sample taken from the upstream and one from the downstream location. Water samples were filtered through 0.45 μm filter membranes (Pall, Ann Arbor, MI, USA) [17,18]. Next the filtered membranes were submerged in 10 mL sterile saline and washed three times [19]. The liquid products were then combined and mixed. The mixture was homogenized and transferred to 10 mL Luria–Bertani (LB) broth (Binhe Corporation, Hangzhou, China) for enrichment. All samples were inoculated onto China Blue agar plates containing 0.3 μg/ml meropenem and incubated overnight in 37 °C. A total of up to 30 colonies was picked per sample from the agar plates [20]. Bacterial species identification was performed using the VITEK® 2 Compact System (Biomérieux Corporation, Paris, France) and Matrix-assisted laser desorption/ionization-time of flight (MALDI-TOF) mass spectrometry (Bruker Corporation, Bremen, Germany). The presence of carbapenem resistance genes (CRGs) was screened by NG-Test® CARBA 5 (Zhongshengzhongjie Bio-Technology Co., Ltd., Changsha, China).

Antimicrobial susceptibility testing

Minimal inhibitory concentrations (MICs) were determined using the broth microdilution technique [21]. The panel of antibiotics tested included imipenem, meropenem, ertapenem, cefmetazole, ceftazidime, cefotaxime, piperacillin/tazobactam, cefoperazone/sulbactam, ceftazidime/avibactam, cefepime, polymyxin B, tigecycline, ciprofloxacin, amikacin, and aztreonam (Luoxin Pharmaceutical, Shandong, China). Interpretation of tigecycline susceptibility referred to the standard of the European Committee on Antimicrobial Susceptibility Testing (EUCAST) [22], while the remaining antibiotics were assessed based on the standard set by the Clinical and Laboratory Standards Institute (CLSI) [23]. *K. pneumoniae* ATCC 25955 was used as a quality control.

Whole-genome sequencing and de novo assembly

Genomic DNA of CRKP isolates was extracted using the Magen Bacterial Genomic DNA extraction kit according to the manufacturer's instructions. Whole-genome sequencing was performed using the Illumina® NovaSeq 6000 platform (Magenbio Biotech Corporation, Shanghai, China). De novo genome assembly was performed using SPAdes version 3.15.1 [24].

Screening Klebsiella pneumoniae strains in retrospective data

For global water source isolates genomic study, we initiated a comprehensive search within the NCBI database to identify all available entries related to "water", "river", "carbapenemases", and "*Klebsiella pneumoniae*" (as of 20 March 2024). This targeted approach allowed us to gather a diverse set of genomic data relevant to our investigation. Following this, we randomly selected 49 strains of carbapenem-susceptible *Klebsiella pneumoniae* (CSKP) from the NCBI database to function as control strains in our analysis.

To investigate the genetic evolutionary background of CRKP in Southeast Asia, we conducted a comprehensive search in the NCBI database using the combined keywords "carbapenem resistance", "Southeast Asia" and "*Klebsiella pneumoniae*" This search yielded 609 CRKP isolates of Southeast Asian origin, encompassing strains from clinical, environmental, and animal sources.

Bioinformatics analysis

Single nucleotide polymorphisms (SNPs) were identified using Parsnp v1.2 of strain N339 as reference [25]. Isolates differing by ≤ 10 allelic differences were considered to constitute a single clone [26]. The species identity, multilocus sequence typing (MLST) and identification of ARGs of isolates were conducted with Kleborate [27]. By comparing the seven housekeeping genes in all the strains, sequence types (STs) were determined. The seven conserved housekeeping genes were obtained from the public databases for molecular typing and microbial genome diversity [28]. Plasmid replicons were identified by PlasmidFinder [29]. Virulence factors were identified based on the core dataset in the Virulence Factor Database (VFDB) [30]. The genomic islands were predicted by IslandViewer 4, and ISfinder 1.0 was used to screen insertion sequences (ISs) [31]. Phylogenetic trees were constructed using Roary and FastTree, based on SNPs in core genomes [32,33]. The phylogenetic tree was visualized and modified using iTOL version 6.5.7 [34].

Galleria mellonella infection model

Larvae of the wax moth (*Galleria mellonella*) [35], each approximately 300 mg, were sourced from Tianjin Huiyude Biotech Company, China, and stored on wood chips in darkness at a temperature of 15 °C until utilization. *K. pneumoniae* strains from overnight cultures were cleaned with phosphate-buffered saline (PBS) and diluted to a standard concentration of 1×10^8 CFU/mL using PBS. Infection of the *G. mellonella* larvae was performed following established protocols, monitoring the survival rate across eight larvae per group [36]. For comparative analysis, a highly virulent ST11 CR-hvKp strain, HVKP4, and a low virulence strain with a deletion in the virulence gene, FJ8, were included [37]. Each experiment was conducted three times to ensure reliability. Kaplan-Meier survival curves were plotted using GraphPad Prism v.7.00 (GraphPad Software Inc., La Jolla, CA, USA).

Statistical analysis

Statistical analysis was conducted by performing the χ^2 test or Fisher's exact test via the IBM SPSS software (IBM, Armonk, USA); $P < 0.05$ was considered as statistically significant. Spearman correlation analysis was used to determine the correlation among ARGs and plasmid replicons, and visualized using Gephi v.0.10.1.

Result

Overview of prevalence of carbapenem-resistant Klebsiella spp. Isolates in China and Nepal

The majority of the isolates detected in this study were *K. pneumonia* (n = 161), followed by *Klebsiella michiganensis* (n = 15), *Klebsiella variicola* (n = 14), and *Klebsiella quasipneumoniae* (n = 2). Among the 161 CRKP isolates, 123 (76.4 %) isolates were recovered from Nepal, including Karmanasha river (n = 13), Bishnumati river (n = 18), Hospital A as a hospital without a wastewater treatment facility (n = 72), and Hospital B as a hospital equipped with a treatment facility wastewater (n = 20). No isolates were isolated from Hospital C, where wastewater is processed through a multi-step treatment system. 38 (23.6 %) strains were recovered from Anhui (n = 35) and Shanghai (n = 3) in China, distributed in 5/105 river sampling sites of 2/10 different provinces. No CRKP was isolated from the samples in Sri Lanka. The highest prevalence of CRKP was 92.1 % (95 % CI, 77.52–97.54 %) in the Nanfei River basin of Hefei city, Anhui Province, which is located two kilometers from four hospitals. In Nepal, CRKP strains were identified in four rivers at eight sampling sites. Among these, the prevalence of CRKP in the untreated discharge of the Hospital A was the highest, at 58.5 % (95 % CI 49.30–67.24 %). The second highest prevalence was observed in the effluent and influent of Hospital B wastewater, with a prevalence of 16.3 % (95 % CI 10.45–24.24 %). There were significant differences in the positive rate of CRKP between the three countries. The CR-*Klebsiella* spp. positive rate at different sampling sites in Nepal (5/8) was significantly higher than that in China (4/105) and Sri Lanka (0/5) ($P < 0.001$). The number of samples collected from different sources and the prevalence of CR-*Klebsiella* spp. strains from different types of samples are shown in Table 1, **Table S1** and Fig. 1.

Profiling resistance phenotypes of the isolates

The antimicrobial susceptibility testing indicated that all of the isolates (n = 192, 100 %) exhibited MDR phenotype and were highly resistant to the antimicrobial agents. Notably, the resistance to ertapenem (ETP), cefmetazole (CMZ), ceftazidime (CAZ), cefepime (FEP), ciprofloxacin(CIP), cefoperazone/sulbactam (SCF) and cefo-

Table 1
Basic information of CRKP isolates collected from river and hospital wastewater.

Types of Samples	Location	Discription	No. of CRKP strains isolated (prevalence%)
Urban aquatic	Qiu Jiang (Shang Hai)	1KM from the First Affiliated Hospital of Naval Medical University (Shanghai Changhai Hospital)	3 (3/38, 7.9 %)
	Nanfeihe River (He Fei)	A tributary of the Yangtze River basin	35 (35/38, 92.1 %)
	Bishnumati river (Katmandu)	2KM upstream and downstream of the Jana River near Hospital B	18 (18/123, 14.6 %)
	Karmanasha river (Katmandu)	2KM upstream and downstream of the Jana River near Hospital A	13 (13/123, 10.6 %)
	Polonnaruwa (Sri Lanka)	7.94′94″ N, 81.01′43″ E	0 (0.0 %)
	Kurunegala (Sri Lanka)	7.48′05″ N, 80.35′72″ E	
	Kandy (Sri Lanka)	7.27′06″ N, 80.60′23″ E	
Drinking water source lake	surface water	27°33′55″ N, 85°37′15″ E	0 (0.0 %)
	Tap water		
	Lake water	Sundarijal National Park	
hospital wastewater	Hospital C (Katmandu)	Inffluent and Effluent	0 (0.0 %)
	Hospital B (Katmandu)		20 (20/123, 16.3 %)
	Hospital A (Katmandu)	Untreated Discharge	72 (72/123, 58.5 %)

taxime (CTX) was universal in China and Nepal, with 100 % resistance rate for all seven antibiotics. Polymyxin B (PB) and tigecycline (TGC) showed no resistance in any of the isolates from China and Nepal. Notably, the MIC results of the 192 isolates from China and Nepal revealed significant differences in resistance to four drugs between strains from the two countries. The isolates from Nepal had significantly higher resistance rates to CAV, AK (amikacin) and TZP (piperacillin/tazobactam) than China ($P < 0.05$). However, for CMZ, the resistance rate of Nepal isolates was lower than that of China ($P < 0.05$) (**Fig. S1,** Table 2).

Distribution characteristics of different carbapenem resistance genes among the isolates

The WGS analysis of 192 isolates showed that 9 CRGs were detected from this study, including *bla*$_{\text{IMP-4}}$ (8.8 %), *bla*$_{\text{KPC-2}}$ (10.4 %), *bla*$_{\text{OXA-181}}$ (2.6 %), *bla*$_{\text{OXA-232}}$ (24.9 %), *bla*$_{\text{OXA-10}}$ (3.1 %), *bla*$_{\text{NDM-1}}$ (57.8 %), *bla*$_{\text{NDM-4}}$ (1.6 %), *bla*$_{\text{NDM-5}}$ (15.1 %) and *bla*$_{\text{NDM-6}}$ (3.1 %). A variety of CRGs were found in these isolates, among which the acquired *bla*$_{\text{NDM-1}}$ and intrinsic *bla*$_{\text{OXA-232}}$ genes were key determinants of carbapenem resistance, with a detection rate of *bla*$_{\text{NDM}}$ approaching 77.6 % (149/192). The distribution of the CRGs showed an obvious geographical specificity, *bla*$_{\text{IMP-4}}$、*bla*$_{\text{NDM-4}}$ and *bla*$_{\text{NDM-6}}$ were only detected in China, *bla*$_{\text{OXA-10}}$、*bla*$_{\text{OXA-181}}$ and *bla*$_{\text{OXA-232}}$ were only detected in Nepal. By comparing the five common CRGs in clinical practice (**Fig. S2a, b**), we found that the carrying rate of *bla*$_{\text{KPC-2}}$ in isolates in China (23.0 %) was significantly higher than that in Nepal (5.0 %) ($P < 0.05$), and the carrying rate of *bla*$_{\text{NDM-1}}$ gene in Nepal (93/132, 70.0 %) was significantly higher than that in China (18/60, 30.0 %) ($P < 0.05$).

In addition, this study compared the co-carriage of CRGs between CR-*Klebsiella* spp. in Nepal and China. Co-carriage of CRGs occurred in both countries, but with significant differences (Fig. S2c). Nepal CR-*Klebsiella* spp. strains carried (39/132, 29.5 %) higher rate of CRGs than that of China (8/60, 13.3 %). In Nepal, 6 types of co-carried CRGs were observed among the 39 Nepalese CR-*Klebsiella* spp. isolates, which were *bla*$_{\text{NDM-5}}$ and *bla*$_{\text{OXA-232}}$ co-carried strains (12/39, 30.8 %), *bla*$_{\text{NDM-5}}$ and *bla*$_{\text{OXA-181}}$ co-carried strains (4/39, 10.3 %), *bla*$_{\text{NDM-5}}$ and *bla*$_{\text{OXA-10}}$ co-carried strains (1/39, 2.6 %), respectively. The percentages of *bla*$_{\text{NDM-1}}$ and *bla*$_{\text{OXA-232}}$ co-carrying strains, *bla*$_{\text{NDM-1}}$ and *bla*$_{\text{KPC-2}}$ co-carrying strains, *bla*$_{\text{NDM-1}}$ and *bla*$_{\text{OXA-10}}$ co-carring strains were 43.6 % (17/39), 10.3 % (4/39), and 2.6 % (1/39), respectively. In China, *bla*$_{\text{KPC-2}}$ and *bla*$_{\text{NDM-1}}$ were the main CRGs co-carriers (5/8, 62.5 %), followed by *bla*$_{\text{IMP-4}}$ and *bla*$_{\text{KPC-2}}$ co-carrying strains (2/8, 25.0 %), and one isolate co-carrying *bla*$_{\text{IMP-4}}$, *bla*$_{\text{KPC-2}}$ and *bla*$_{\text{NDM-1}}$, which is unique.

To explore the relationship between ARGs and plasmid replicons, a correlation analysis was conducted (Fig. 2). We considered a correlation significant when the correlation coefficient R value was > 0.3 and $P < 0.05$. In the intra-group correlation analysis of ARGs in CRKP isolates of China, *bla*$_{\text{IMP-4}}$ showed a strong positive correlation with *aac(6′)-Ib* and *aadA1* (R value > 0.6). On the other hand, *bla*$_{\text{NDM-5}}$ exhibited a negative correlation with *bla*$_{\text{NDM-1}}$ and *qnrS1* (R value < -0.4). In contrast, in CRKP isolates from Nepal, *bla*$_{\text{NDM-1}}$ had a strong positive correlation with *qnrS1* (R value > 0.8) and a negative correlation with *aadA2* (R value < -0.8). The inter-group correlation analysis between ARGs and plasmid replicons showed that, in CRKP isolates of China, *bla*$_{\text{IMP-4}}$ had a strong positive correlation with IncFIA and IncFII(p14) (R value = 1). In CRKP isolates from Nepal, *bla*$_{\text{NDM-1}}$ was positively correlated with IncN (R value > 0.5).

Genetic contexts of blaNDM-1 in the isolates

The genetic environment of *bla*$_{\text{NDM-1}}$ analysis revealed the typically conserved gene order in our study as *hp-ble*$_{\text{MBL}}$-*bla*$_{\text{NDM-1}}$. BLASTN analysis further confirmed that the *bla*$_{\text{NDM-1}}$ gene in different bacterial strains, such as *Providencia rettgeri* strain P50213 and *Serratia marcescens* strain S50079, consistently resides within a conserved genomic context, flanked by a hypothetical protein-coding gene and the bleomycin-binding protein gene *ble*$_{\text{MBL}}$. In additional, the *bla*$_{\text{NDM-1}}$ cassette of No.925 included genes for chloramphenicol CatB-family (*catB*), subclass B1 β-lactamase (*bla*$_{\text{NDM-1}}$), bleomycin resistance protein (Ble-MBL), phosphoribosylanthranilate isomerase (PRAI), and IS*CR1* family transposase (IS*91*-like). Besides, the *bla*$_{\text{NDM-1}}$ cassette exhibited a high degree of similarity to those identified in *P. rettgeri* and *S. marcescens* strain (**Fig. S3**).

Phenotypic characteristics of the recovered Klebsiella spp

WGS analysis was conducted to comprehensively characterize the genetic features of the selected cohort, comprising 192 isolates. In total, 18 ARGs and six types of mutations were characterized. Aside from the carbapenem resistance genes, these ARGs collectively conferred resistance to aminoglycosides (*aadA, armA, rmtB, strA, and aac*), fosfomycin (*fosA3*), fluoroquinolone (*qnrB, qnrS*),

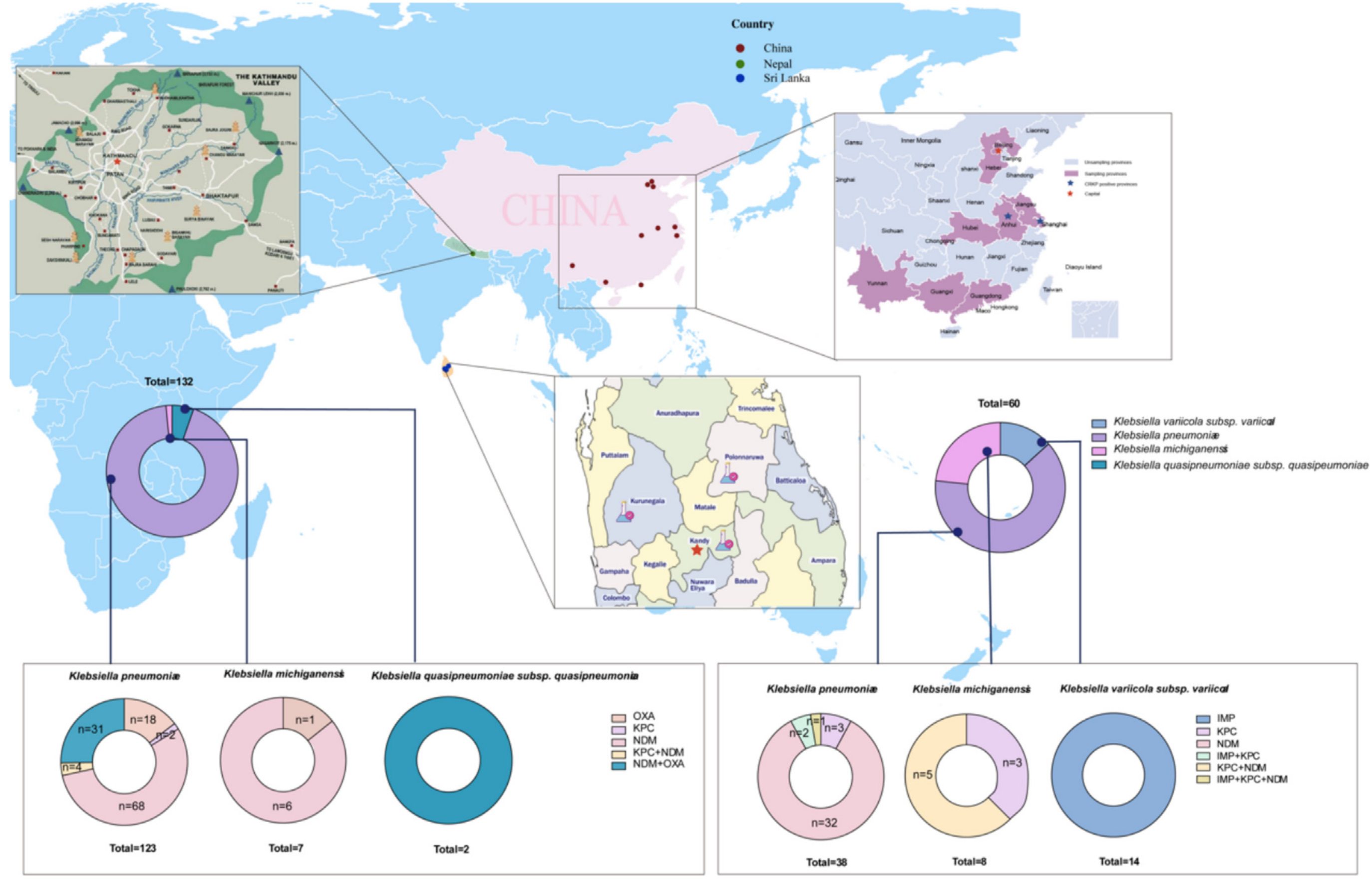

Fig. 1. Locations of sampling cities and their prevalence. Regions highlighted with purple points denote cities from which the samples were collected. The majority of isolates detected in this study were *K. pneumonia* (n = 161), followed by *K. michiganensis* (n = 15), *K. variicola* (n = 14), and *K. quasipneumoniae* (n = 2). A circular diagram was used to represent the distribution of carbapenemases across the different *Klebsiella* species, with each color indicating a distinct carbapenemase type.

Table 2
MIC result of isolates isolated from China and Nepal to 15 Antimicrobial agents.

Antimicrobials classes	Antimicrobial agents	CHINA				NEPAL			
		MIC90	MIC50	R%	Total	MIC90	MIC50	R%	Total
Carbapenems	IPM	8	8	100.00	60	128	8	68.94	91
	MEM	32	8	100.00	60	128	16	68.94	91
	ETP	16	16	100.00	60	>128	16	100.00	132
Cephalosporins	CMZ	>128	>128	76.67	46	>128	32	45.45	60
	CAZ	>128	>128	100.00	60	>128	>128	100.00	132
	CTX	>128	>128	100.00	60	>128	>128	100.00	132
	FEP	>64/4	32	100.00	60	>64	64	100.00	132
β-lactam combination agents	TZP	>128	>128	76.67	46	>256/4	>256/4	100.00	132
	SCF	>256/4	>256/4	100.00	60	>256/128	>256/128	100.00	132
	CAV	>256/128	>256/128	70.00	42	>64/4	>64/4	83.33	110
Polymyxins	PB	<0.5	<0.5	0.00	0	2	<0.5	0.00	0
Tetracycline	TGC	1	0.5	0.00	0	2	<0.25	0.00	0
Fluoroquinolones	CIP	16	2	100.00	60	>32	16	100.00	132
Aminoglycosides	AK	<2	<2	0.00	0	>64	64	48.48	64
Monobactams	ATM	>128	32	73.33	44	>128	>128	66.67	88

Abbreviations: IMP, imipenem; MEM, meropenem; ETP, ertapenem; CMZ, cefmetazole; CAZ, ceftazidime; CTX, cefotaxime; TZP, piperacillin/tazobactam; SCF, cefoperazone/sulbactam; CAV, ceftazidime/avibactam; FEP, cefepime; PB, polymyxin B; TGC, tigecycline; CIP, ciprofloxacin; AK, amikacin; ATM, aztreonam.

Phenicols (*cmlA*, *catA*, and *catB*), rifampicin (*arr-2*, *arr-3*), sulfonamide (*sul1*, *sul2*), tetracyclines (*tet*(A), *tet*(B), *tet*(C), *tet*(D), and *tet*(R)), tigecycline (*tmexC1D1-toprJ1*), trimethoprim (*dfrA*), and beta-lactamase extended-spectrum beta-lactamase (bla_{CTX-M}, bla_{OXY}, bla_{SFO}, and bla_{SHV}). Thses findings are consistent with the multiple antimicrobial resistance phenotypes previously observed in these isolates (**Table S2**).

Genetic diversity of CRKP strains in the Southeast Asian countries

Based on NCBI database and our statistics, a phylogenetic tree was constructed encompassing 609 CRKP isolates from 5 countries of Southeast Asia (SEA) (365 from China, 180 from Nepal, 45 from Thailand, 14 from Cambodia and 5 from Singapore) and mapped against the acquired ARGs, key virulence factors, MRG metal(loid)

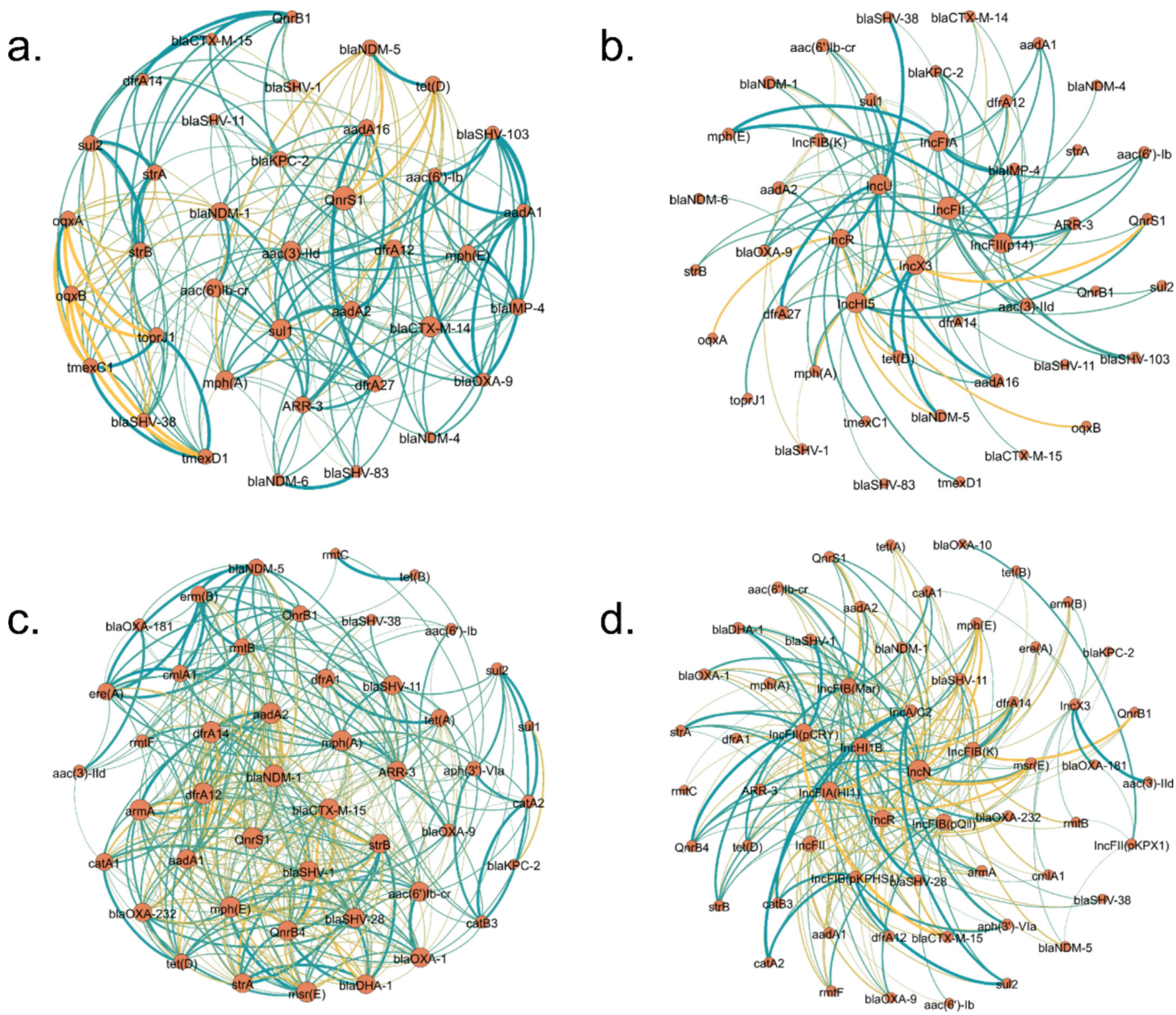

Fig. 2. The network graph describing the co-occurrence pattern of ARGs with plasmid replicons. a-b, Intra-group correlations among ARGs in CRKP isolates of China and inter-group correlations between ARGs and plasmid replicons. c-d, Intra-group correlations among ARGs in CRKP isolates of Nepal and inter-group correlations between ARGs and plasmid replicons. Connections between nodes indicate that they are related. Green colors and thicker lines represent stronger positive correlations. The yellow color depth on the line shows the strength of the negative correlation. Moreover, the line's thickness is proportional to the correlation strength, with a thicker line indicating a stronger relationship between the variables. All associated genes in the figure had *P* values less than 0.05.

resistance genes, chlorine resistance genes, and plasmid replicon types (Fig. 3a). Based on the NCBI database (n = 448) and isolates in this study (n = 161), the $bla_{\text{NDM-1}}$ gene was identified in a broad range of 143 clinical and environmental isolates. Among the 143 isolates, 98 isolates were from China and Nepal River. 23 isolates were obtained from human, mouse and unknown host within China, 19 isolates were obtained from Nepal human, and a smaller subset of 3 isolates from Thailand. The presence of the $bla_{\text{KPC-2}}$ gene was observed in 91 CRKP isolates, with a geographical bias towards samples from China (n = 85) and Nepal (n = 6). The predominance of $bla_{\text{KPC-2}}$ in human samples from China (50/85). In contrast, the gene was detected exclusively in environmental strains from river water in Nepal. The $bla_{\text{NDM-5}}$ gene also showed a restricted distribution, being identified only in homo, chicken, and river samples from China and in river strains from Nepal.

In addition, the quinolone resistance genes (*qnrA, qnrB, qnrS*) was detected in 65.9 % (482/732) of all strains. This high prevalence was particularly pronounced in river water samples, with the Nepalese river water isolates showing a prevalence rate of 83.3 % (110/132). In China, the river water isolates showing a carrying rate of 80.0 % (48/60), indicating a widespread distribution of this resistance gene in aquatic environments. In addition, 2.4 % (18/732) of the CRKP strains from China and Thailand were found to carry the *tet*(X) gene, which was not detected in our study isolates. The tigecycline-resistant efflux pump gene, *tmexCD-toprJ*, was identified in 4.5 % (33/732) of CRKP strains originating from homo, meat (chicken, duck and pig) and water sources in China. A significant proportion of CRKP strains, 61.8 % (453/733), were found to carry the sulfonamide resistance gene (*sul1*, *sul2* and *sul3*). The cyclolide resistance genes (*msrE* and *mphG*) were detected in 34.4 % (252/733) of CRKP strains.

The plasmid replicon and sequence types of CRKP isolates

We conducted a WGS analysis of the 733 CRKP isolates and identified their plasmid replicon types. The results showed that

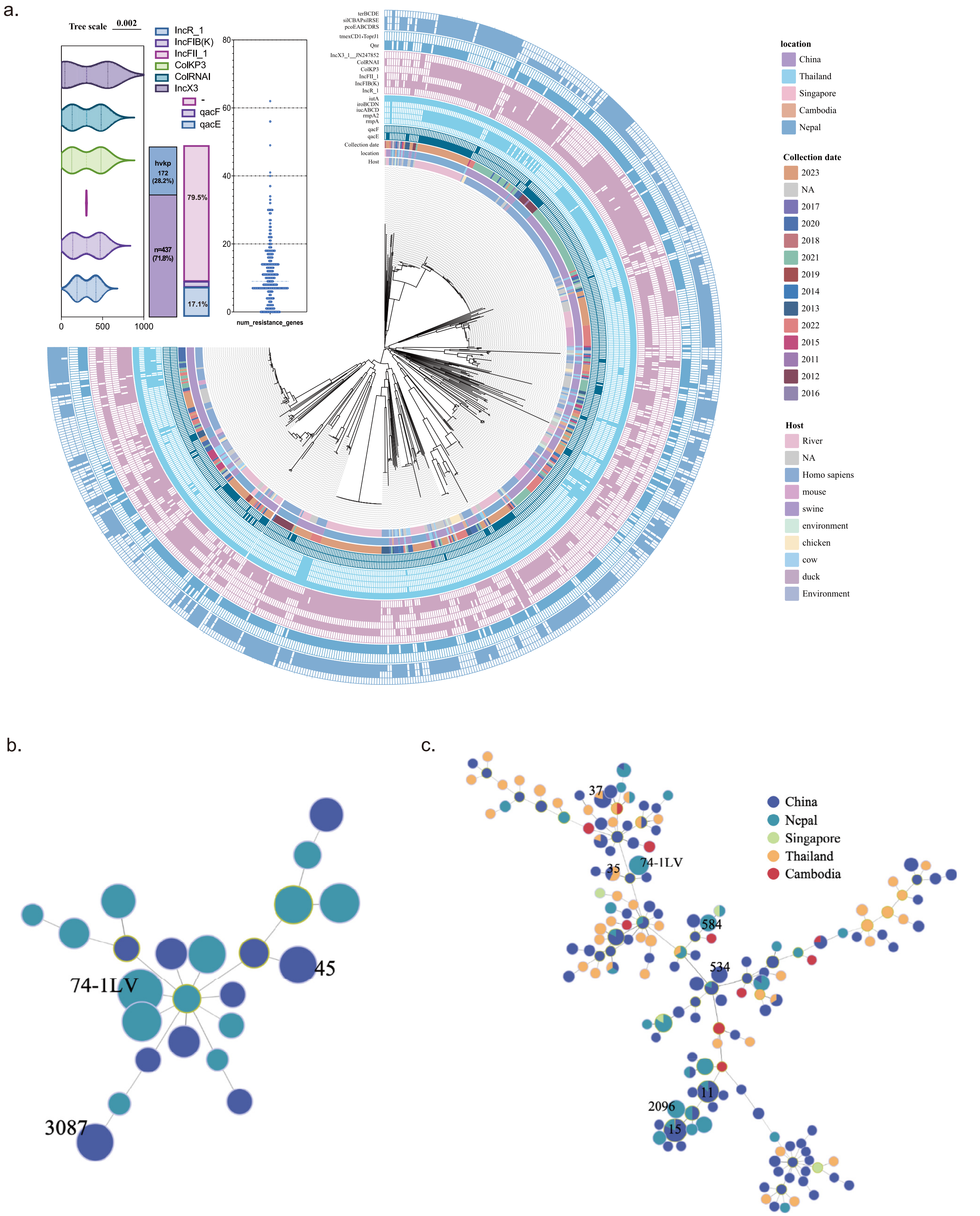

Fig. 3. Phylogeny and Minimal spanning tree of CRKP. a. Phylogenetic tree of CRKP isolates from Southeast Asian countries. Source, host, plasmids, ARGs, virulence resistance genes, chlorine disinfection tolerance gene (*qacE*), and heavy metal resistance gene (MRG) were annotated. Branches of strains belonging to our study were labelled with red colors. b. STs distribution of CRKP isolates in this study. c. STs distribution of CRKP isolates in Southeast Asian countries.

the CRKP strains carried a variety of incompatibility (Inc) plasmids, including IncFIB (76.4 %), IncFII (50.4 %) and ColRNAI (63.0 %). Among the 192 CRKP strains in this study, IncFIB plasmids were detected in China (58/60, 96.6 %) and Nepal (108/132, 81.2 %). Meanwhile, IncFII plasmids were detected in China (19/60, 41.7 %) and Nepal (100/132, 75.8 %). Our results show a higher detection frequency of CRKP isolates with IncFIB plasmids in China compared to Nepal, in contrast to the predominance of isolates with IncFII in Nepal.

Among the Nepal 132 CRKP strains, 126 could be categorized into 10 STs, with seven belonging to a novel ST type. The most prevalent ST was ST74-1LV (39/132, 29.5 %). In China, 52 could be categorized into 9 STs, with 8 strains belonging to a novel ST. The most prevalent STs were ST45 (14/60, 23.3 %) and ST3087 (14/60, 23.3 %) (Fig. 3b). Remarkably, among 733 CRKP isolates, we identified a diverse array of 147 STs. The top three most common STs were ST15 (13.2 %), ST11 (9.0 %), and ST45 (5.0 %), followed by ST147 (2.9 %), ST2096 (2.7 %), ST37 (2.3 %), ST25 (2.0 %), ST584(2.0 %), and ST3392(2.0 %). Additionally, there were 136 other STs, each represented by 1–14 isolates (Fig. 3c). Notably, ST15 (n = 97) and ST11 (n = 66) were the most common genotypes found in patients with clinical in China, while ST2096 (n = 12), ST584 (n = 12), ST3087 (n = 14), and ST45 (n = 17) were prevalent in river and environment.

Characterization of quaternary ammonium compound resistantce (QACs) and metal(loid) resistance (MGR) genes

In order to estimate the percentage of QACs genes in isolates, three QAC resistance genes (*qacE, qacEΔ1, and qacF*) were used to screen all isolates. The results indicated that 36.8 % (270/733) and 2.9 % (21/733) of the strains contained the *qacE* and *qacF* resistance genes, respectively. Among the different sources of CRKP strains in five countries, *qacE* was detected only in China river (25/60, 41.7 %), Nepal river (70/133, 52.6 %) and homo samples (21/60, 35.0 %).

We also detected several genes encoding resistance to heavy metals, including copper (*pcoABCDRS*), silver (*silABCEPSR*), and tellurium (*terABCDEFZ*) in all isolates. The *silABCEPSR* gene was identified in 53.9 % (395/733) of all strains. The tellurium (*terABCDEFZ*) gene was detected in 33.4 % (245/733) of the strains.

Global genomic epidemiology of CRKP isolates from aquatic sources

To further investigate the global genomic epidemiological distribution and evolutionary characteristics of CRKP from water sources, we downloaded the genomes of global waterborne CRKP isolates (n = 42) and selected genomes of various ST CSKP isolates as control strains (n = 49) (Fig. 4a). This anlysis has identified the global presence of 203 CRKP in water sources across eight distinct countries. These isolates were obtained from various aquatic environments in Australia, Brazil, the United States, Portugal, Bangladesh, China, Nepal, and Germany, highlighting the widespread distribution of this antibiotic-resistant pathogen (Fig. 4b). A phylogenetic tree based on core genome SNPs was constructed. In our analysis of CRGs among the 203 isolates, we identified a total of 10 distinct CRGs. The predominant genes detected were $bla_{\text{NDM-1}}$ (10/203, 49.8 %), $bla_{\text{OXA-48-like}}$ (55/203, 27.1 %), and $bla_{\text{NDM-5}}$ (34/203, 16.7 %). While the bla_{KPC} gene has been previously reported as dominant in clinical *K. pneumoniae* isolates, our findings reveal that bla_{NDM} is the predominant carbapenemase gene identified in water sources. The maximum likelihood phylogeny built from single nucleotide polymorphisms (SNPs) extracted from 252 core genes revealed a genetically diverse population. We identified 65 known sequence types. The ST diversity is high. Only nine STs contained at least ten genomes (ST11, 13 genomes; ST147, 23 genomes; ST2096, 19 genomes; ST528, 10 genomes; ST45, ST258, ST584 and ST3392, 15 genomes each; ST74-1LV, 39 genomes. A total of 88 and 56 STs were represented by less than eight genome(s), respectively.

A total of 35 waterborne *K. pneumoniae* strains were identified as harboring the virulence genes *iuc1/3* or *rmpA2*. Among these, 34 strains, classified as CR-hvKp, were isolated from Nepalese water sources. Additionally, one strain of hvKp from Chinese water samples was identified to contain the virulence gene *iuc3*. The analysis of all *K. pneumoniae* 45 serotypes and discovered that KL64 (44/252), KL111 (39/252), and KL10 (32/252) were the predominant strains. 93.2 % (41/44) of the KL64 serotype isolates were identified as CRKP. Furthermore, KL64-CR-hvKp strains represented 75.6 % (31/41) of all KL64-CRKP strains. The majority of KL64-CR-hvKp isolates belonged to ST3392 (15/34, 44.1 %) and ST2096 (19/34, 55.9 %), with both KL64-CR-hvKp isolates predominantly isolated from Nepal. Additionally, we found the ST11-KL64-CRKP strain harboring the $bla_{\text{KPC-2}}$ gene in six CRKP isolates from Brazil. However, no virulence genes were detected in the six isolates from Brazil.

Utilizing G. mellonella larvae as a model to investigate the virulence of CR- Klebsiella spp. from China and Nepal

Genomic analysis revealed that 19.6 % (144/733) isolates possessed at least two siderophores biosynthesis genes: *iuc* (aerobactin) and the hypermucoid regulator *rmpA/rmpA2* gene (**Table S3**). The most prevalent virulence factors *iutA* (encoding and regulating with aerobactin) 18 were detected at 99.5 % (729/733). It is noteworthy that the virulence factors of CRKP strains from Chinese and Nepalese rivers exhibited notable discrepancies ($p < 0.05$). The *rmpA2* and *iuc* genes were identified in 25.8 % (34/132) of CRKP strains from Nepalese rivers, whereas no associated virulence factors were detected in 60 CRKP strains from Chinese rivers. According to the analysis of virulence genes revealed that Nepal isolates possessed the plasmid-encoded siderophores biosynthesis genes: *iuc* (aerobactin) and the hypermucoid regulator *rmpA/rmpA2* gene. *G. mellonella* larvae was used to assess the virulence of this isolate, and the larvae were infected. Plasmid HVKP4 was used as a hyper-virulence control and FJ8 as a low-virulence control. The results indicated that isolates N339, N320, N381, and N751 from Nepal caused mortality in *G. mellonella* larvae at a concentration of 10^7 CFU/mL, exhibiting similar virulence to the hypervirulence control strain HVKP4. In contrast, the Chinese isolates 631 and 633 exhibited similar virulence to the lower-virulence control strain FJ8 (Figs. 5a, b).

Discussion

Microbial profile and contaminants are the most detrimental parameters reducing the drinking water quality [38,39]. The genomic analysis of carbapenem-resistant- *Klebsiella* spp. isolates from urban aquatic environments in China and Nepal provides a stark illustration of the epidemiological and evolutionary challenges posed by antibiotic resistance. The stark contrast in prevalence between China (31.2 %) and Nepal (68.8 %) suggests a complex interplay of healthcare infrastructure, environmental management, and antibiotic use policies [40]. The higher prevalence in Nepal could be attributed to various factors, including differences in antibiotic stewardship, environmental conditions, and population dynamics.

The identification of different CRGs, particularly the predominance of $bla_{\text{NDM-1}}$, aligns with global trends and indicates the rapid spread of these resistance determinants beyond hospital settings. The geographical specificity of certain CRGs, such as $bla_{\text{IMP-4}}$ and

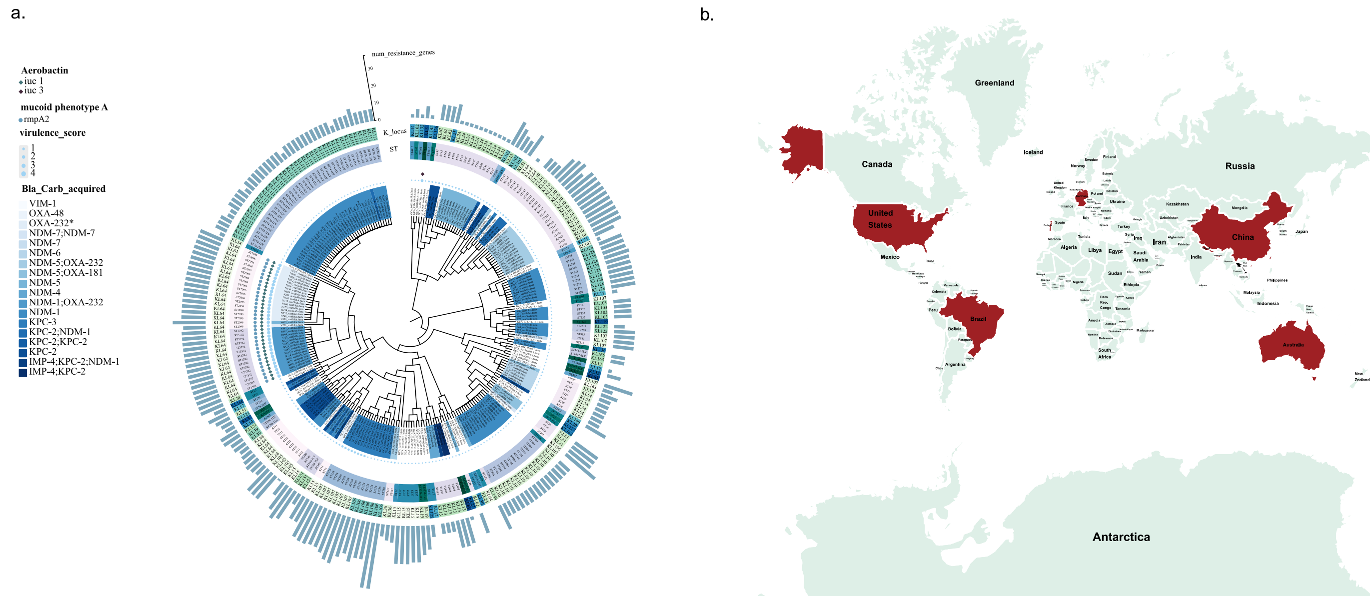

Fig. 4. Phylogeny tree of CRKP in global water . Source. a. Phylogenetic tree of CRKP isolates from water of global countries. Key characteristics including KL serotype, sequence types (STs), the count of antimicrobial resistance genes (ARGs), presence of virulence factor genes, and the class of carbapenem-resistance determinants are annotated. Branches of carbapenem-susceptible strains download from NCBI were labelled with none colors. b. Mapping of global distribution of carbapenem-resistant *Klebsiella pneumoniae* isolates from aquatic environments as observed up to the year 2024

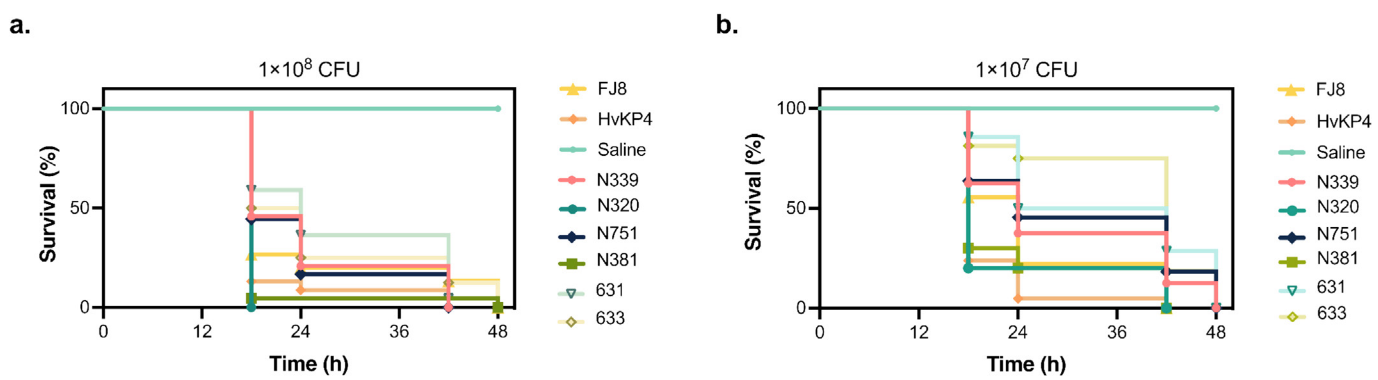

Fig. 5. Virulence potential of CRKP isolates (N339, N320, N751, N381, 631, and 633) using a *G. mellonella* infection model The horizontal axis of the graph represents the survival time of *G. mellonella* larvae and the vertical axis indicates the survival percentage (%).

bla$_{OXA-232}$, provides a nuanced perspective on the local evolution and transfer of resistance elements. The co-occurrence of CRGs and their different prevalence rates between the two countries imply different evolutionary pressures.

The identification of quaternary ammonium compound resistance and MGR genes in CRKP introduces a significant challenge to clinical and agricultural practices [41,42]. Traditionally, chlorine-containing disinfectants and heavy metals have been mainstays in clinical disinfection and agricultural applications, respectively. However, the emergence of CRKP strains harboring these resistance genes could potentially restrict the availability of effective clinical antimicrobial agents and complicate the management of CRKP infections, emphasizing an urgent need to explore alternative strategies for infection control and environmental stewardship.

Phylogenetic analysis and identification of different STs among CRKP isolates reveal a complex landscape of clonal expansion and diversification. The high prevalence of certain STs, such as ST15 and ST11, in clinical settings and the environmental association of others, such as ST2096 and ST3087, suggest a dynamic interplay between human, animal and environmental reservoirs. The widespread dissemination of CRKP in aquatic environments, as evidenced by our genomic epidemiological analysis of waterborne isolates, raises significant concerns regarding its potential for global spread. Isolates identified across eight countries highlight the role of water sources as key vectors in the transmission of CRKP beyond healthcare settings.

The presence of CRKP in such diverse aquatic ecosystems underscores the ubiquity of this resistant pathogen and suggests that it may be gaining a foothold in environments beyond clinical settings. The predominance of *bla*$_{NDM}$ in environmental isolates contrasts with clinical settings where *bla*$_{KPC}$ is often reported as the dominant carbapenemase [39]. This discrepancy could indicate a differential selection pressure exerted by environmental conditions or variations in antibiotic usage patterns between clinical and environmental spheres. The genetic diversity observed within the CRKP population, with 65 known sequence types identified, is indicative of a complex evolutionary landscape. The high ST diversity and the presence of a few dominant STs, such as ST11, ST147, ST2096, ST528, ST45, ST258, ST584, and ST3392, suggest multiple successful lineages that have potentially adapted to environmental pressures. The finding that KL64, KL111, and KL10 are the predominant serotypes among the isolates further emphasizes the global distribution of specific CRKP clones.

The detection of virulence factors in waterborne strains, particularly the presence of *iuc*1/3 or *rmpA2* in 34 Nepalese CR-hvKp isolates, is concerning. In addition, our study identified a noteworthy occurrence of the ST11-KL64-CRKP strain carrying the bla_{KPC-2} gene in Brazilian isolates, even in the absence of detected virulence genes. This finding underscores the potential for CRKP strains to disseminate globally through the acquisition of specific resistance genes, independent of virulence factors. Drawing from other research, the ST11-KL64 CRKP sublineage is reported to have acquired a pK2044-like virulence plasmid from the hypervirulent ST23-KL1 *K. pneumoniae* [43]. The pK2044-like plasmid demonstrates a strong host adaptability to ST11-KL64-CRKP, leading us to hypothesize that aquatic environments may play a pivotal role in the evolutionary trajectory of CRKP into CR-hvKp, representing a dynamic shift in the microbial population. Using *G. mellonella* as an infection model, we observed differences in virulence between Chinese and Nepalese isolates that correlated with the presence of specific virulence genes, highlighting the multifactorial nature of pathogenicity in *Klebsiella* spp..

China's advanced wastewater treatment practices in both clinical and environmental sectors have mitigated the spread of resistant strains into the environment [44]. In contrast, Nepal's healthcare facilities, while serving a vital role in a country known for its rich cultural and religious heritage [45], may inadvertently contribute to the dissemination of resistance genes through inadequate medical wastewater management [46]. The direct discharge of untreated or insufficiently treated medical wastewater into aquatic systems poses a significant risk for the spread of resistance to the community and poses a substantial challenge to public health [47]. Our clinical focus reveals that the environmental impact of antimicrobial resistance is a critical area of concern. The conditionally management of hospital wastewater in China has resulted in a reduced dissemination of resistant bacteria and resistance genes into the external environment. Conversely, Nepal's healthcare facilities face a heightened risk of contributing to environmental contamination, which may, in turn, exacerbate the spread of resistance.

While the study provides valuable genomic data on CRKP, limitations include the potential for selection bias in sampling sites and the need for larger-scale, longitudinal studies to better understand the dynamics of resistance gene dissemination. Future research should also focus on the mechanisms of resistance evolution, the role of mobile genetic elements in horizontal gene transfer, and the development of novel therapeutic strategies to combat multidrug resistance.

Conclusion

Our study provides a comprehensive analysis of the epidemiological distribution and genomic characteristics of CRKP in urban aquatic environments across multiple countries, offering an in-depth overview of its resistance, virulence, and evolutionary patterns. We identified significant regional differences in antimicrobial resistance profiles and ARGs compositions among CRKP isolates. Additionally, our findings underscore the pivotal role of mobile genetic elements in ARG dissemination. Notably, the detection of CR-hvKp in water samples raises serious public health concerns. These results underscore the urgent need for global surveillance and effective environmental management strategies to mitigate the spread of CRKP in aquatic ecosystems.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jare.2025.09.020.

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